

Enhanced Attack Report

Umbrella Stand

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Definitions

Pre-Conditions: Conditions that must be true to execute the attack steps in the milestone.

Post-Conditions: Traces that an attacker leaves behind after executing the attack steps in the milestone.

Attack Steps: Steps that an attacker would take to achieve the goal of the milestone.

MITRE Technique: Techniques from the MITRE ATT&CK; framework that are relevant to the milestone.

STIX

Malware Name: Umbrella Stand

Malware Description: UMBRELLA STAND beacons to its C2 server using fake TLS on port 443. UMBRELLA STAND has functionality to run shell commands. UMBRELLA STAND has been observed targeting FortiGate 100D series firewalls produced by Fortinet. UMBRELLA STAND has command configurable C2 and beacon frequency. UMBRELLA STAND has been observed deployed in conjunction with a set of publicly available tooling including: BusyBox, nbtscan, tcpdump and openLDAP. UMBRELLA STAND is able to hook the reboot functionality of the device.

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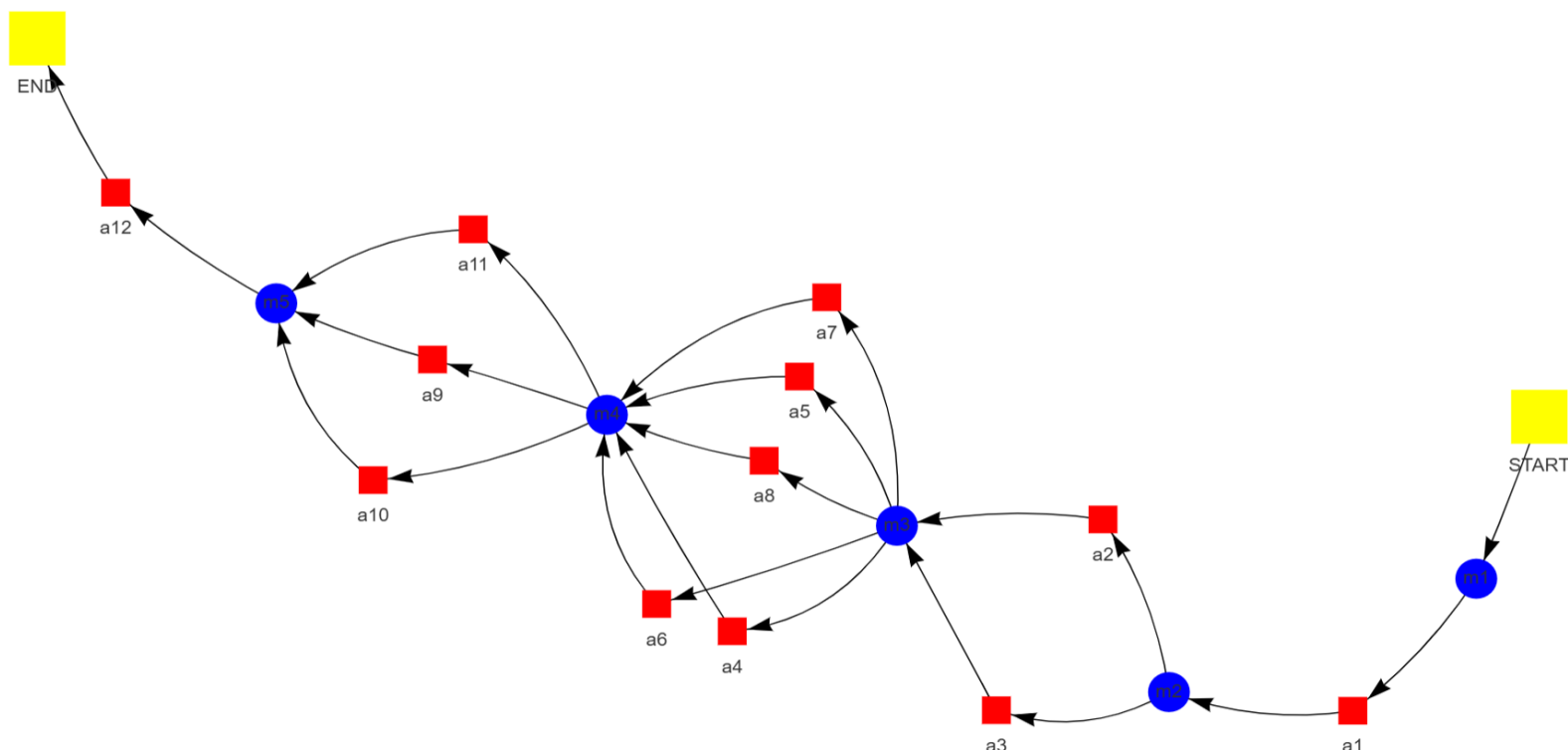
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Attack Graph



Milestone *m1*

Attack Step *a1*

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Name: Command and Scripting Interpreter: Unix Shell as used by the malware

Description: UMBRELLA STAND malware facilitates remote command execution (RCE) on compromised FortiGate firewalls through the utilization of Unix shells such as BusyBox or ash. This functionality enables attackers to remotely manipulate files, modify network configurations, and gather system information from the compromised device. Connection establishment between the attacker and UMBRELLA STAND is likely achieved via a TLS-based channel, potentially obscuring malicious activity. Commands are transmitted through this encrypted channel for execution by UMBRELLA STAND using either BusyBox or ash shells. Subsequently, the results of executed commands are returned to the attacker. The inherent capability of arbitrary command execution grants attackers complete control over the compromised firewall, posing significant security implications. These include data exfiltration, distributed denial-of-service (DDoS) attacks, and lateral movement within the network. Mitigation strategies encompass maintaining up-to-date FortiGate firmware to address known vulnerabilities, implementing network segmentation to limit breach impact, deploying intrusion detection/prevention systems (IDS/IPS) for suspicious activity monitoring, and enforcing multi-factor authentication (MFA) for administrative access.

Pre-Conditions:

- The malware has gained sufficient privileges to execute shell commands.
- The malware has successfully infected a system with a Unix-based operating system.

Post-Conditions:

- New user accounts or permissions changes in system logs.
- Increased risk of data breaches and system instability.
- Remote access to the compromised device by the attacker.
- Data exfiltration from the network.
- Potential for further malware deployment and lateral movement within the network.
- Modified firewall configuration files.
- Presence of UMBRELLA STAND binaries and related files on the device.
- Compromised FortiGate firewall functionality.
- Encrypted data files with specific encryption patterns.
- Unusual network traffic patterns, including connections to known malicious IP addresses.
- Deleted log entries or altered timestamps indicating attempts to cover tracks.

MITRE Technique

ID: T1059.004

Name: Command and scripting interpreter: unix shell

Description: Adversaries may abuse Unix shell commands and scripts for execution. Unix shells are the primary command prompt on Linux, macOS, and ESXi systems, though many variations of the Unix shell exist (e.g. sh, ash, bash, zsh, etc.) depending on the specific OS or distribution. Unix shells can control every aspect of a system, with certain commands requiring elevated privileges.

More info: <https://attack.mitre.org/techniques/T1059/004>

Milestone *m2*

Attack Step a2

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Name: Boot or Logon Autostart Execution: Kernel Modules and Extensions as used by the malware

Description: Boot or logon autostart execution is achieved by UMBRELLA STAND through kernel module and extension modification techniques. A component designated as "reboot_hooker" interfaces with the system's reboot process, enabling the execution of initialization code ("cisz") upon device restart. This mechanism ensures persistence even after a reboot event.

Pre-Conditions:

- The target device is running a Linux operating system.
- The reboot_hooker file exists and is accessible.
- The SYSV664564856 file exists and is accessible.

Post-Conditions:

- Exfiltrated data files transferred from the compromised device.
- Logs indicating unusual activity and command execution.
- Modified configuration files related to firewall rules and security settings.
- 44.194.32 on port 443 using fake TLS traffic.

MITRE Technique

ID: T1547

Name: Boot or logon autostart execution

Description: Adversaries may configure system settings to automatically execute a program during system boot or logon to maintain persistence or gain higher-level privileges on compromised systems. Operating systems may have mechanisms for automatically running a program on system boot or account logon. These mechanisms may include automatically executing programs that are placed in specially designated directories or are referenced by repositories that store configuration information, such as the Windows Registry. An adversary may achieve the same goal by modifying or extending features of the kernel.

More info: <https://attack.mitre.org/techniques/T1547/>

Attack Step a3

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Name: Hijack Execution Flow: Dynamic Linker Hijacking as used by the malware

Description: Dynamic linker hijacking (ldpreload) is employed by the UMBRELLA STAND malware for persistence and execution flow manipulation. The ldpreload mechanism in Linux permits the specification of shared libraries (.so files) to be loaded into every process initiated on the system prior to any other library loading. This ensures early execution of a custom library. Upon program initiation, the dynamic linker (ld) searches for and loads shared libraries specified by the LD_PRELOAD environment variable. UMBRELLA STAND injects its malicious library (likely named libguic.so) into /etc/ld.so.preload, ensuring automatic loading of libguic.so by all processes on the compromised system. Due to early loading, libguic.so can modify system behavior by altering functions called by

legitimate programs, redirecting output or injecting malicious code. It can also hook critical functions, such as network connections and file access, for attacker manipulation. The libguic.so library likely contains malware components and communication channels with the attacker's command-and-control (C2) server. ldpreload's system-wide impact affects all processes, making detection and removal challenging. The malicious code operates within a seemingly legitimate context, blending in with system functions. Persistence is ensured by the ldpreload mechanism, maintaining malware activity even after reboots.

Pre-Conditions:

- The target system is running a Linux operating system.
- The malware has the ability to create and inject its own malicious code into the memory space of the targeted application.
- The malware has identified a vulnerable application that utilizes dynamic linking.
- The malware has successfully gained access to the target system.
- The malware possesses the capability to modify system files, specifically the shared library paths.

Post-Conditions:

- Presence of UMBRELLA STAND binaries and associated files.
- Lateral movement within the victim's network.
- Modified .ini files containing task instructions and collected data.
- Traces of command execution using BusyBox, nbtscan, tcpdump, and openLDAP.
- Hidden files and directories in "/data2/.ztlsl/".
- Potential for disruption of network services.
- Altered SSH configurations and reverse SSH tunnels.
- Logs indicating unusual activity, including process creation, file modifications, and network connections.
- Network traffic analysis revealing encrypted communications on port 443 without proper TLS handshakes.
- Long-term unauthorized access to the Fortinet FortiGate firewall.
- Data exfiltration from the compromised network.
- Modified "sysctl" binary with altered directory references.

MITRE Technique

ID: T1574.001

Name: Hijack execution flow: dll

Description: Adversaries may abuse dynamic-link library files (DLLs) in order to achieve persistence, escalate privileges, and evade defenses. DLLs are libraries that contain code and data that can be simultaneously utilized by multiple programs. While DLLs are not malicious by nature, they can be abused through mechanisms such as side-loading, hijacking search order, and phantom DLL hijacking.

More info: <https://attack.mitre.org/techniques/T1574/001>

Milestone *m3*

Attack Step *a4*

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Name: Deobfuscate/Decode Files or Information as used by the malware

Description: UMBRELLA STAND's deobfuscation techniques are characterized by the extensive use of AES encryption for safeguarding sensitive data, including embedded strings within the malware code itself. This encryption methodology presents a significant barrier to direct analysis by security tools. Despite employing encrypted communication with command and control (C2) servers, UMBRELLA STAND utilizes hardcoded IP addresses, such as 89.44.194.32. This suggests a potential limitation in the number of known C2 servers, which could be subject to updates by the attackers over time. The deobfuscation process is likely initiated with the decryption of core malware components, including encrypted strings and potentially other data structures. This decryption would necessitate the utilization of a specific AES key possessed exclusively by the attackers. Subsequent to decryption, meaningful strings are reconstructed from the scrambled data. These reconstructed strings may encompass commands intended for execution on the infected system, network addresses (including C2 servers), and configuration settings governing the malware's behavior. The primary challenge in deobfuscation efforts centers around the recovery of the AES key employed by UMBRELLA STAND. Without this key, decrypting the malware payload and comprehending its full functionality proves extremely difficult. Furthermore, malware frequently employs techniques such as code injection or dynamic code loading to further obfuscate its actions, rendering static analysis (analysis conducted without execution) less effective. Security researchers leverage a range of tools and techniques for deobfuscation purposes. Disassemblers and decompilers facilitate the breakdown of machine code into more readily interpretable assembly language, aiding in the identification of patterns and potential decryption routines. Hex editors are utilized to examine the raw binary data of the malware, searching for suspicious patterns or strings that may indicate encryption algorithms. Dynamic analysis sandboxes provide controlled environments for executing malware, enabling researchers to observe its behavior and network interactions to uncover deobfuscation techniques. It is imperative to approach malware deobfuscation with caution and adherence to best practices for malware analysis to mitigate the risks of accidental infection or data compromise.

Pre-Conditions:

- Tools capable of reversing engineering and analyzing malware code.
- The target system contains files or data encoded by the malware.
- Access to the infected system containing the obfuscated files.
- The malware utilizes obfuscation techniques to hide its true functionality.
- Potential need for specialized decryption keys or algorithms if the data is encrypted.

Post-Conditions:

- Remote access by attacker
- Altered process names
- Unusual file access patterns
- Hidden directories and files
- System instability
- Loss of sensitive information
- Compromised system functionality
- Potential for further attacks

- Log entries indicating suspicious activity
- Data exfiltration
- Presence of malicious binaries
- Modified SSH configurations
- Encrypted data with specific IVs
- Modified system files
- Network connections to C2 server

MITRE Technique

ID: T1140

Name: Deobfuscate/decode files or information

Description: Adversaries may use Obfuscated Files or Information to hide artifacts of an intrusion from analysis. They may require separate mechanisms to decode or deobfuscate that information depending on how they intend to use it. Methods for doing that include built-in functionality of malware or by using utilities present on the system.

More info: <https://attack.mitre.org/techniques/T1140/>

Attack Step a5

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Name: Process Injection as used by the malware

Description: Dynamic linker hijacking is employed by UMBRELLA STAND for process injection purposes. This technique involves manipulation of the system's dynamic linking mechanism to facilitate the loading of a malicious library into a legitimate process. Target processes are identified, with "usbmux" being a likely candidate based on observations. The malware injects its own malicious library, designated as "libguic.so", into the target process's address space. This is accomplished through modification of the system's LD_PRELOAD environment variable. Upon program initiation, the LD_PRELOAD variable is consulted for libraries to be loaded prior to the program's own. By setting LD_PRELOAD to point to "libguic.so", UMBRELLA STAND ensures its library is loaded first. Upon loading, the injected "libguic.so" library executes code that verifies if the process is indeed "usbmux". If confirmed, a specific initialization component referred to as "cisz" is executed within the context of the legitimate "usbmux" process. This effectively enables UMBRELLA STAND to execute its malicious payload without raising suspicion, as it operates disguised within a seemingly benign process.

Pre-Conditions:

- The dskz process injection tool is available within the UMBRELLA STAND malware suite.
- The target system is running a vulnerable version of FortiGate 100D series firewall.
- UMBRELLA STAND malware has successfully compromised the system.

Post-Conditions:

- Increased risk of further attacks and data breaches
- Compromised Fortinet firewall functionality
- Potential for lateral movement within the network
- Data exfiltration from the compromised device
- Modified system files (e.g., /bin/sysctl, reboot hooks)
- Altered SSH configurations
- Log entries indicating unusual activity or process injections
- Presence of malware binaries (e.g., blghtd, jvnlp, cisz)

- Network traffic to the C2 server on port 443 with fake TLS headers
- Modified system registry entries related to ldpreload techniques
- Remote control of the compromised device
- Hidden directories containing malware files

MITRE Technique

ID: T1055.002

Name: Process injection: portable executable injection

Description: Adversaries may inject portable executables (PE) into processes in order to evade process-based defenses as well as possibly elevate privileges. PE injection is a method of executing arbitrary code in the address space of a separate live process.

More info: <https://attack.mitre.org/techniques/T1055/002>

Attack Step a6

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Name: Indicator Removal: File Deletion as used by the malware

Description: File deletion by the UMBRELLA STAND malware is inferred to be executed through the utilization of system calls or libraries. Operating system functions such as unlink(), or analogous equivalents across diverse systems, are commonly employed by malware for direct file removal. Alternatively, UMBRELLA STAND may leverage higher-level libraries or tools designed for file manipulation, potentially obscuring the underlying system calls.

Pre-Conditions:

- The malware is running on a system with file deletion capabilities.

Post-Conditions:

- Modified reboot function in the Fortinet operating system.
- Hidden directories containing UMBRELLA STAND files and associated tooling.
- "libguic.so" library loaded into new processes.
- Logs indicating process injection and execution of malicious commands.
- Potential for further attacks and lateral movement within the network.
- Loss of system integrity and confidentiality.
- Deleted UMBRELLA STAND binaries from the device.
- Traces of tcpdump and openLDAP usage.
- Compromised Fortinet firewall system.
- Altered SSH configurations allowing reverse SSH connections.
- Network traffic between the compromised system and the C2 server, encrypted using AES in CBC mode.
- Data exfiltration from the compromised system.
- Modified "/bin/sysctl" binary redirecting references to protected directories.
- Remote access to the compromised system granted to attackers.

MITRE Technique

ID: T1027.005

Name: Obfuscated files or information: indicator removal from tools

Description: Adversaries may remove indicators from tools if they believe their malicious tool was detected, quarantined, or otherwise curtailed. They can modify the tool by removing the indicator and

using the updated version that is no longer detected by the target's defensive systems or subsequent targets that may use similar systems.

More info: <https://attack.mitre.org/techniques/T1027/005>

Attack Step a7

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Name: Hide Artifacts: Process Argument Spoofing as used by the malware

Description: Process argument spoofing is employed by UMBRELLA STAND malware to conceal its malicious activities. This technique involves the modification of process identifiers to resemble legitimate system processes. The malware analyzes active processes on the infected system, selecting common and innocuous names for disguise. Subsequently, UMBRELLA STAND utilizes system calls or API functions to alter its own process name and potentially other process attributes, such as descriptions, to align with the chosen legitimate process name. This deceptive practice enables UMBRELLA STAND to evade detection by security tools reliant on blacklisting known malware names and to gain user trust by appearing as a benign process.

Pre-Conditions:

- The target device is running a Fortinet FortiGate 100D series firewall.
- The sysctl binary is available and functional on the target device.
- The malware has successfully compromised the device and gained control.

Post-Conditions:

- 44.194.32).
- Files containing stolen data or configuration information.
- Traces of public domain utilities (BusyBox, nbtscan, tcpdump) used by UMBRELLA STAND.

MITRE Technique

ID: T1564.010

Name: Hide artifacts: process argument spoofing

Description: Adversaries may attempt to hide process command-line arguments by overwriting process memory. Process command-line arguments are stored in the process environment block (PEB), a data structure used by Windows to store various information about/used by a process. The PEB includes the process command-line arguments that are referenced when executing the process. When a process is created, defensive tools/sensors that monitor process creations may retrieve the process arguments from the PEB.

More info: <https://attack.mitre.org/techniques/T1564/010>

Attack Step a8

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Name: Hide Artifacts: Hidden Files and Directories as used by the malware

Description: Hidden files and directories are employed by UMBRELLA STAND and its associated tooling to conceal their presence on infected devices. The ".ztl" directory, for instance, serves as a repository for malware binaries and configuration files. Its leading "." character renders it hidden from standard view in most operating systems. Files utilized by UMBRELLA STAND frequently adhere to cryptic or non-descriptive naming conventions, further obscuring their purpose and functionality. The

creation of hidden directories and files is facilitated through standard file system permissions mechanisms. By setting specific access control flags, such as the "hidden" attribute, these files and directories become invisible to users unless explicit system configuration reveals them. This tactic effectively hinders detection by security tools and analysts who rely on visible file listings for identifying malicious activity. Consequently, UMBRELLA STAND is able to operate stealthily, evading scrutiny and maintaining persistence on compromised devices.

Pre-Conditions:

- The malware has the capability to create hidden files and directories.

Post-Conditions:

- Altered network configurations allowing for outbound connections to the C2 server.
- Compromised system with persistent remote access for attackers.
- Network traffic analysis revealing encrypted communication with the C2 server on port 443 without proper TLS handshakes.
- Modified system registry entries related to process execution and file access.
- Modified system files, including hooks in reboot functionality.
- Presence of UMBRELLA STAND binaries and associated tooling (e.g., BusyBox, nbtscan, tcpdump).
- Data exfiltration from the compromised device.
- New user accounts or modified existing ones with elevated privileges.
- Increased risk of further attacks and malware infections.
- Potential for lateral movement within the network.
- Traces of deleted files or modified timestamps suggesting attempts to cover tracks.
- Logs indicating unusual activity, including shell command executions and data transfers.

MITRE Technique

ID: T1564.001

Name: Hide artifacts: hidden files and directories

Description: Adversaries may set files and directories to be hidden to evade detection mechanisms. To prevent normal users from accidentally changing special files on a system, most operating systems have the concept of a 'hidden' file. These files don't show up when a user browses the file system with a GUI or when using normal commands on the command line. Users must explicitly ask to show the hidden files either via a series of Graphical User Interface (GUI) prompts or with command line switches (dir /a for Windows and ls -la for Linux and macOS).

More info: <https://attack.mitre.org/techniques/T1564/001>

Milestone *m4*

Attack Step a9

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Name: Process Discovery as used by the malware

Description: Process discovery within the UMBRELLA STAND malware framework is facilitated by a watchdog mechanism designed to ensure the continuous operation of its primary process, designated as "blghtd." This mechanism likely entails periodic system-level checks to ascertain the status and existence of the "blghtd" process. System calls or libraries such as "ps" are potentially employed to retrieve a comprehensive list of active processes. Subsequently, a comparison is conducted between the process ID (PID) of the "blghtd" process and the retrieved list. Based on the outcome of this comparison, appropriate actions are taken. If the "blghtd" process is identified as operational, the watchdog mechanism persists, potentially involving timer resets or other maintenance procedures to ensure continued process execution. Conversely, if the "blghtd" process is absent, corrective measures may be initiated, such as self-restart or attempts to reestablish its presence within the system environment.

Pre-Conditions:

- The malware is running on a target system.
- The UMBRELLA STAND malware is present and active on the target system.
- The target system has processes executing.
- The target system has network connectivity to its command and control server.

Post-Conditions:

- Increased vulnerability to further attacks.
- New processes running with unusual names (e.g., blghtd, cisz, jvnlp).
- Hidden directories containing malicious files (/data2/.ztl/).
- Log entries indicating unauthorized access and file modifications.
- Altered system registry settings.
- Presence of encrypted data files with unusual naming conventions.
- System instability and performance degradation.
- Unusual network traffic patterns (e.g., beaconing on port 443, reverse SSH connections).
- Data exfiltration and/or manipulation.
- Compromised system with persistent backdoor access.
- Modified system files (e.g., /bin/sysctl, SSH configurations).

MITRE Technique

ID: T1057

Name: Process discovery

Description: Adversaries may attempt to get information about running processes on a system. Information obtained could be used to gain an understanding of common software/applications running on systems within the network. Administrator or otherwise elevated access may provide better process details. Adversaries may use the information from Process Discovery during automated discovery to shape follow-on behaviors, including whether or not the adversary fully infects the target and/or attempts specific actions.

More info: <https://attack.mitre.org/techniques/T1057/>

Attack Step a10

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Name: Network Service Discovery as used by the malware

Description: Network service discovery is conducted by UMBRELLA STAND through the utilization of the open-source tool nbtscan. Deployment of nbtscan occurs on compromised Fortinet devices. Subsequently, configured and executed instances of nbtscan scan the local network for NetBIOS computer names. Active computers are identified through the transmission of NetBIOS name queries and analysis of corresponding responses. Information gathered may include: potential targets within the network; a mapping of the network topology; and intelligence regarding operating systems and applications running on other devices. Discovered information is likely transmitted to a UMBRELLA STAND command-and-control server for analysis and the formulation of further exploitation strategies.

Pre-Conditions:

- Network connectivity to potential target systems.
- The device has an active network connection.
- The "nbtscan" tool is available and functional within the malware's environment.
- The malware is running on a Fortinet FortiGate 100D series firewall.
- A running instance of the UMBRELLA STAND malware.

Post-Conditions:

- Presence of UMBRELLA STAND malware files and artifacts on the device.
- Modified ini files containing beaconing intervals and C2 redirection information.
- Shell command execution logs.
- 44.194.32).
- Network traffic captured using tools like tcpdump and nbtscan.

MITRE Technique

ID: T1016.001

Name: System network configuration discovery: internet connection discovery

Description: Adversaries may check for Internet connectivity on compromised systems. This may be performed during automated discovery and can be accomplished in numerous ways such as using Ping, tracer, and GET requests to websites.

More info: <https://attack.mitre.org/techniques/T1016/001>

Attack Step a11

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Name: Network Sniffing as used by the malware

Description: The utilization of tcpdump by UMBRELLA STAND for network sniffing involves several stages. tcpdump is likely installed on the compromised FortiGate firewall through exploitation of vulnerabilities or legitimate file transfer mechanisms. Subsequently, UMBRELLA STAND configures tcpdump to capture specific network traffic based on predefined rules encompassing protocols, IP addresses/hostnames, and port numbers. Continuous monitoring of network interfaces by tcpdump results in the capture of packets adhering to the established rules. Captured packets are either stored in a file or processed in real-time by UMBRELLA STAND for analysis. This analysis may entail

extraction of sensitive information from unencrypted traffic, identification of communication patterns and system logins, and construction of a network map. Finally, captured data may be exfiltrated from the compromised FortiGate firewall to an attacker-controlled server for further examination and exploitation.

Pre-Conditions:

- The malware has access to the network interface of the target device.
- The target device is connected to a network.

Post-Conditions:

- Potential for further attacks and lateral movement within the network.
- Loss of system integrity and confidentiality.
- Presence of "libguic.so" library loaded into new processes.
- Network traffic logs showing communication with the C2 server over port 443 using fake TLS headers.
- Modified /etc/ld.so.preload file.
- Modified reboot functionality on the device.
- Log entries indicating unusual activity, process injections, and network connections to the C2 server.
- New processes running with suspicious names or functionalities.
- Traces of file manipulation and system modifications.
- Encrypted files with a specific pattern (AES-CBC encryption with fixed IV).
- Hidden directories containing malware binaries (/data2/.ztlsl/).
- Compromised system with persistent backdoor access.
- Altered sysctl utility configurations.
- Data exfiltration from the infected device.

MITRE Technique

ID: T1040

Name: Network sniffing

Description: Adversaries may passively sniff network traffic to capture information about an environment, including authentication material passed over the network. Network sniffing refers to using the network interface on a system to monitor or capture information sent over a wired or wireless connection. An adversary may place a network interface into promiscuous mode to passively access data in transit over the network, or use span ports to capture a larger amount of data.

More info: <https://attack.mitre.org/techniques/T1040/>

Milestone *m5*

Attack Step *a12*

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Name: Data Obfuscation: Protocol Impersonation as used by the malware

Description: Protocol Impersonation is employed by UMBRELLA STAND for data obfuscation and evasion of detection. Network packets are crafted to superficially resemble legitimate TLS connections, encompassing TLS headers, ciphers, and protocol elements. This mimicry conceals malicious intent by exploiting the trust associated with secure communication protocols. While appearing to utilize TLS encryption, the actual data exchange between UMBRELLA STAND and its Command and Control (C2) server may be unencrypted or employ weak encryption methods. This facilitates the transmission of sensitive information such as commands, stolen data, or system details without raising suspicion. The impact of this technique is manifested in the difficulty of detecting UMBRELLA STAND's malicious activity due to: Traditional security solutions reliant on TLS inspection may be misled into perceiving traffic as legitimate, enabling malware bypass detection. Analysis of seemingly encrypted data presents complexities and time constraints, hindering investigators' ability to ascertain the true nature of communication. Mitigation strategies against this threat encompass: Implementation of advanced threat detection mechanisms that transcend basic TLS inspection and incorporate behavioral analysis techniques for identification of suspicious network traffic patterns. Regular monitoring of network activity for unusual connections or data transfers indicative of protocol impersonation attempts. Employee education regarding phishing attacks and social engineering tactics that may lead to malware infections such as UMBRELLA STAND.

Pre-Conditions:

- The malware is running on a target device.
- A target device with network connectivity.
- The malware has established a connection to its Command and Control (C2) server.
- The malware has access to the necessary system resources for protocol impersonation.

Post-Conditions:

- Presence of AES encryption keys used by the malware.
- Encrypted data transfers between infected device and C2 server.
- Modified configuration files related to firewall rules or security settings.
- Traces of executed shell commands in system logs.

MITRE Technique

ID: T1001.003

Name: Data obfuscation: protocol or service impersonation

Description: Adversaries may impersonate legitimate protocols or web service traffic to disguise command and control activity and thwart analysis efforts. By impersonating legitimate protocols or web services, adversaries can make their command and control traffic blend in with legitimate network traffic.

More info: <https://attack.mitre.org/techniques/T1001/003>